

CONSOLE LOGS

```
888b1a10399083d73a/
gs://dataproc-staging-us-central1-94667674826-sjkq5ce5/google-cloud-dataproc-metainfo/6f9e1e88-97af-43c9-b404-42e900034a4d/jobs/9432e519767f4e888b1a10399083d73a/driveroutput.*
tunir bhattacharya@cloudshell:~ (projecttwo-494921)$ gsutil ls gs://amz_dataset/raw/
Google recommends using Gcloud storage CLI (https://docs.cloud.google.com/storage/docs/discover-object-storage-gcloud) instead of gsutil. Please refer to migration guide (https://docs.cl
oud.google.com/storage/docs/gsutil-transition-to-gcloud) for assistance.
CommandException: One or more URLs matched no objects.
tunir bhattacharya@cloudshell:~ (projecttwo-494921)$ ^C
tunir bhattacharya@cloudshell:~ (projecttwo-494921)$ gcloud dataproc jobs submit pyspark amz_dataset.py --cluster=projecttwo --region=us-central1 -- gs://amz_dataset/amz
_ca total_products_data_processed.csv gs://amz_dataset/output/ 2>&1 | tee job_run.log
Job [fd0c96e0349647d48cf9073ff633belb] submitted.
Waiting for job output...
26/04/30 22:31:36 INFO SparkEnv: Registering MapOutputTracker
26/04/30 22:31:36 INFO SparkEnv: Registering BlockManagerMaster
26/04/30 22:31:36 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/04/30 22:31:36 INFO SparkEnv: Registering OutputCommitCoordinator
26/04/30 22:31:37 INFO MetricsConfig: Loaded properties from hadoop-metrics2.properties
26/04/30 22:31:37 INFO MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
26/04/30 22:31:37 INFO MetricsSystemImpl: google-hadoop-file-system metrics system started
26/04/30 22:31:38 INFO DataprocSparkPlugin: Registered 188 driver metrics
26/04/30 22:31:38 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at projecttwo-m.c.projecttwo-494921.internal./10.128.0.3:8032
26/04/30 22:31:38 INFO AHSProxy: Connecting to Application History server at projecttwo-m.c.projecttwo-494921.internal./10.128.0.3:10200
26/04/30 22:31:39 INFO Configuration: resource-types.xml not found
26/04/30 22:31:39 INFO ResourceUtils: Unable to find 'resource-types.xml'.
26/04/30 22:31:40 INFO YarnClientImpl: Submitted application application_1777586229027_0004
26/04/30 22:31:41 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at projecttwo-m.c.projecttwo-494921.internal./10.128.0.3:8030
26/04/30 22:31:43 INFO GoogleCloudStorageImpl: Ignoring exception of type GoogleJsonResponseException; verified object already exists with desired state.
26/04/30 22:31:44 INFO GoogleHadoopOutputStream: hflush(): No-op due to rate limit (RateLimiter[stableRate=0.2qps]): readers will *not* yet see flushed data for gs://dataproc-temp-us-cen
tral1-94667674826-oxsz2zke/6f9e1e88-97af-43c9-b404-42e900034a4d/spark-job-history/application_1777586229027_0004.inprogress [CONTEXT ratelimit_period="1 MINUTES" ]
Spark version: 3.5.3
Reading from: gs://amz_dataset/amz_ca total_products_data_processed.csv
Writing to: gs://amz_dataset/output
Total cleaned rows: 1988016

=== Main comparison table (top 20) ===
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|categoryName|price_bucket|product_count|avg_list_price|avg_current_price|avg_stars|avg_bought_last_month|total_bought_last_month|avg_discount_pct|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Beauty| MID |11042|4.5|18.17|4.21|172.5|1905050|16.36|
|Beauty| LOW |3713|2.36|7.83|4.27|260.0|965250|22.87|
|Grocery| LOW |3681|0.94|5.99|4.05|201.2|740600|24.32|
|Grocery| MID |7202|2.92|19.23|3.94|85.6|616400|17.12|
|Home Kitchen| MID |1794|6.19|19.89|4.39|343.0|615300|17.45|
|Health Personal Care| MID |1311|5.33|20.24|4.43|435.4|570750|16.45|
|Pet Supplies| MID |5181|4.69|19.72|4.2|109.2|565850|15.87|

Writing outputs to GCS...
Pipeline complete.
Job [fd0c96e0349647d48cf9073ff633belb] finished successfully.
done: true
driverControlFilesUri: gs://dataproc-staging-us-central1-94667674826-sjkq5ce5/google-cloud-dataproc-metainfo/6f9e1e88-97af-43c9-b404-42e900034a4d/jobs/fd0c96e0349647d48cf9073ff633belb/
driverOutputResourceUri: gs://dataproc-staging-us-central1-94667674826-sjkq5ce5/google-cloud-dataproc-metainfo/6f9e1e88-97af-43c9-b404-42e900034a4d/jobs/fd0c96e0349647d48cf9073ff633belb/
driveroutput
jobUuid: 50ad54ee-60d9-330b-813b-25c54268ed52
placement:
  clusterName: projecttwo
  clusterUuid: 6f9e1e88-97af-43c9-b404-42e900034a4d
pysparkJob:
  args:
  - gs://amz_dataset/amz_ca total_products_data_processed.csv
  - gs://amz_dataset/output/
  mainPythonFileUri: gs://dataproc-staging-us-central1-94667674826-sjkq5ce5/google-cloud-dataproc-metainfo/6f9e1e88-97af-43c9-b404-42e900034a4d/jobs/fd0c96e0349647d48cf9073ff633belb/stag
ing/amz_dataset.py#1777588287737060
referenceC:
  jobId: fd0c96e0349647d48cf9073ff633belb
  projectId: projecttwo-494921
status:
  state: DONE
  stateStartTime: '2026-04-30T22:33:37.446730Z'
statusHistory:
- state: PENDING
  stateStartTime: '2026-04-30T22:31:27.937752Z'
- state: SETUP_DONE
  stateStartTime: '2026-04-30T22:31:28.035073Z'
- details: Agent reported job success
  state: RUNNING
  stateStartTime: '2026-04-30T22:31:28.494084Z'
yarnApplications:
- memoryMbSeconds: '1203945'
  name: AmazonProductsPipeline_PA5
  progress: 1.0
  state: FINISHED
  trackingUrl: http://projecttwo-m.c.projecttwo-494921.internal.:8080/proxy/application_1777586229027_0004/
  vcoreSeconds: '365'
```

BUCKET FOLDER PROOFS

I>	Buckets > amz__dataset > output					
	Create folder Upload Transfer data Create batch operations New Other services					
	Filter by name prefix only Filter Filter objects and folders					
	☐	Name	Size	Type	Created ?	Storage class Last modified
	☐	category_summary/	—	Folder	—	—
	☐	main_comparison_table/	—	Folder	—	—
	☐	rating_vs_sales/	—	Folder	—	—
	☐	top_sellers_with_context/	—	Folder	—	—

	Create folder Upload Transfer data Create batch operations New					
	Filter by name prefix only Filter Filter objects and folders					
	☐	Name	Size	Type		
	☐	_SUCCESS	0 B	application/octet-stream		
	☐	price_bucket=HIGH/	—	Folder		
	☐	price_bucket=LOW/	—	Folder		
	☐	price_bucket=MID/	—	Folder		
	☐	price_bucket=PREMIUM/	—	Folder		

COMPUTER USAGE PROOFS

